Aura Developer Kit Carrier Board Specification

DX-5000-D

Specification

Document History

Author	Version	Date	Description of Change
HaoNing	1.0	2026.1.2	Document initial
HaoNing	1.1	2026.2.6	Added : Device ordering information
HaoNing	1.2	2026.4.15	Updated: 1.J0901/0902/0801 headers pin definition 2.J1201/J1202 ethernet PHY to 10Gbased 3. Added wakeup/rest key header jumper J2004/J2002
HaoNing	1.2.1	2026.5.9	Updated and corrected technical parameters.
HaoNing	1.2.2	2026.5.22	Updated: 1. Table 19 Supply Current Capabilities per Connector , update current and power capability. 2. Updated and corrected parameters.
HaoNing	1.2.3	2026.6.29	Updated and corrected parameters.
Shichenkang	1.2.4	2026.6.29	1. Converted full-width punctuation to half-width in English text; 2. removed zero-width characters.
HaoNing	1.2.5	2026.7.16	Correction: The J1803 interface type-C only supports USB 2.0

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1 Introduction

This specification contains recommendations and guidelines for engineers to follow to create modules for the expansion connectors on the Aura Module carrier board. As well as understand the capabilities of the other dedicated interface connectors and associated power solutions on the platform.

1.1 Aura Carrier Board Feature List

The following is a list of features for the carrier board.

- Connection to Aura Module
  - 314-pin MxM connector + 12-pin B2B connector.
- USB
  - USB 3.1/2.0
    - 2x USB Type-A USB3.1 (host only)
    - 1x USB Type-C : USB3.1 with OTG
    - 1x USB Type-C : USB2.0 host
  - 1x USB Micro-B : Device programming & debug only
- Wired network
  - 10GBASE-T 2x RJ45 connector with LEDs
- Display
  - 1x HDMI 2.0
  - 1x 22-position flex connectors
    - DSI , D-PHY(V1.2) Support 4 data-lanes per port,2.5Gbps per lane
- Camera connectors
  - 2x 22-position flex connectors
    - CSI , D-PHY(V2.1) 2 Lanes or 4 Lanes,4.5Gbps per lane
    - Camera CLK, I2C, and control
- M.2 Key E connector
  - PCIe (Gen4) x1 Lane
  - I2S, UART
- M.2 Key M connector
  - PCIe (Gen4) x2 Lane, control RC or EP Mode
- Expansion header
  - 30-pin (2x15) header, 26-pin (2x13) header, 20-pin (2x10) header
    - 6x CANFD/ 2x SPI/ 4x PWM/ 1x I2S/ 3x I2C/ 5x UART/ 2x ADC
- UI and indicators
  - Button header: Power, Reset, Wakeup
- Power and Miscellaneous
  - Fan connector: 12V, PWM
  - Power DC Jack: 12-20 V (12V supply provided)

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- Main 5.0 V
- Main 3.3V
- Main 1.8V

➢ Developer kit operating temperature range

0~45℃

1.2 Aura Carrier Board Block Diagram

Figure 1 to Figure 5 show the block diagram and various placement views for Aura core board and the carrier board.

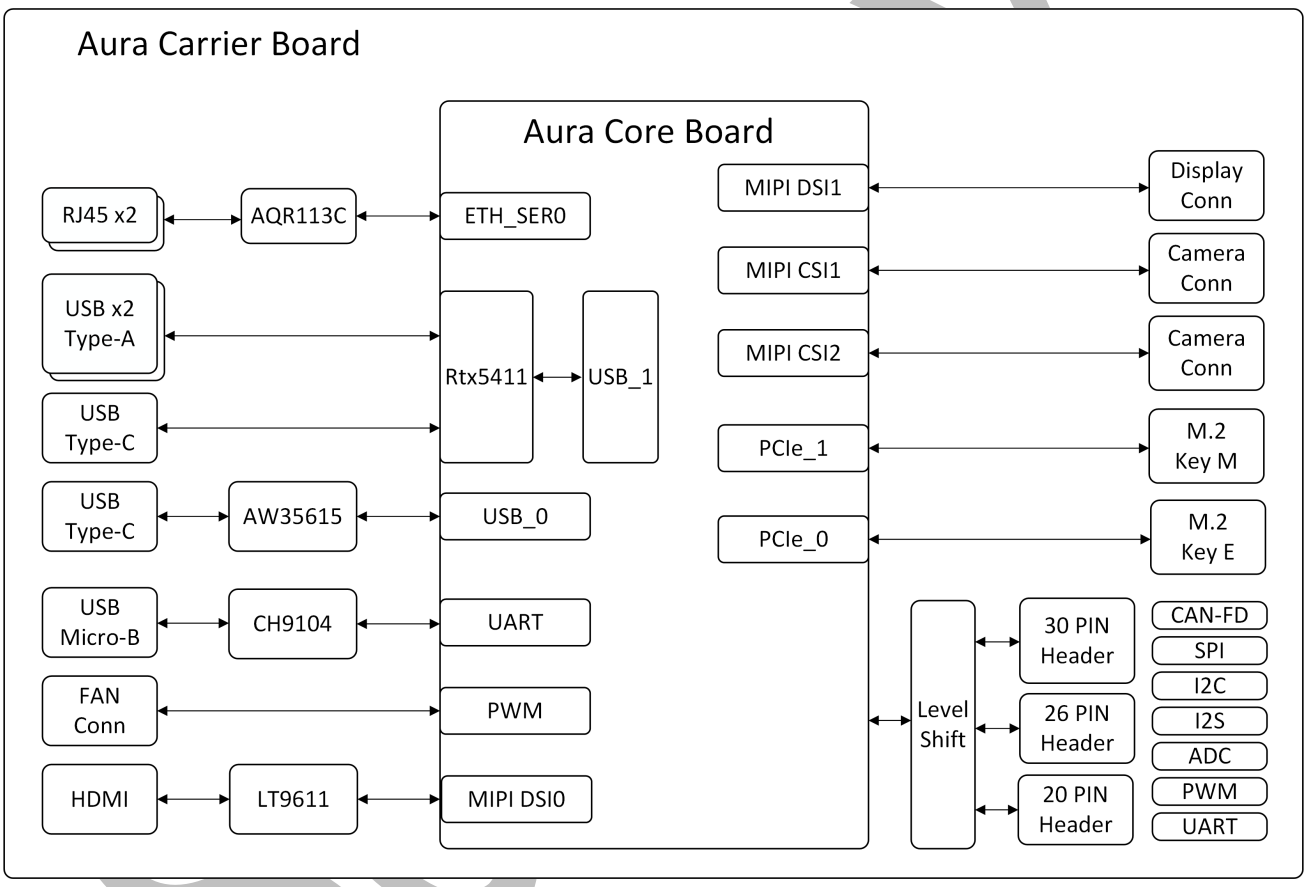


Figure 1 Aura Module Core&Carrier Board Block diagram

2 Mechanical Specification

Unit : mm

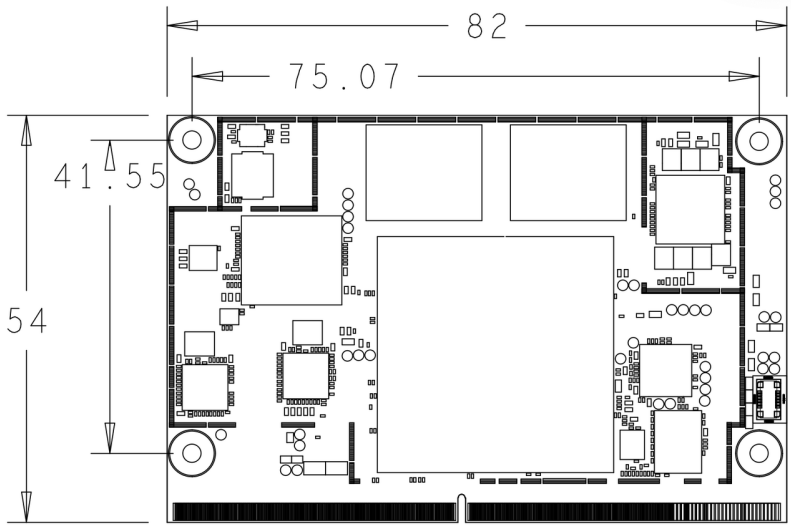


Figure 2. Aura Module Core Board Mechanical Dimensions

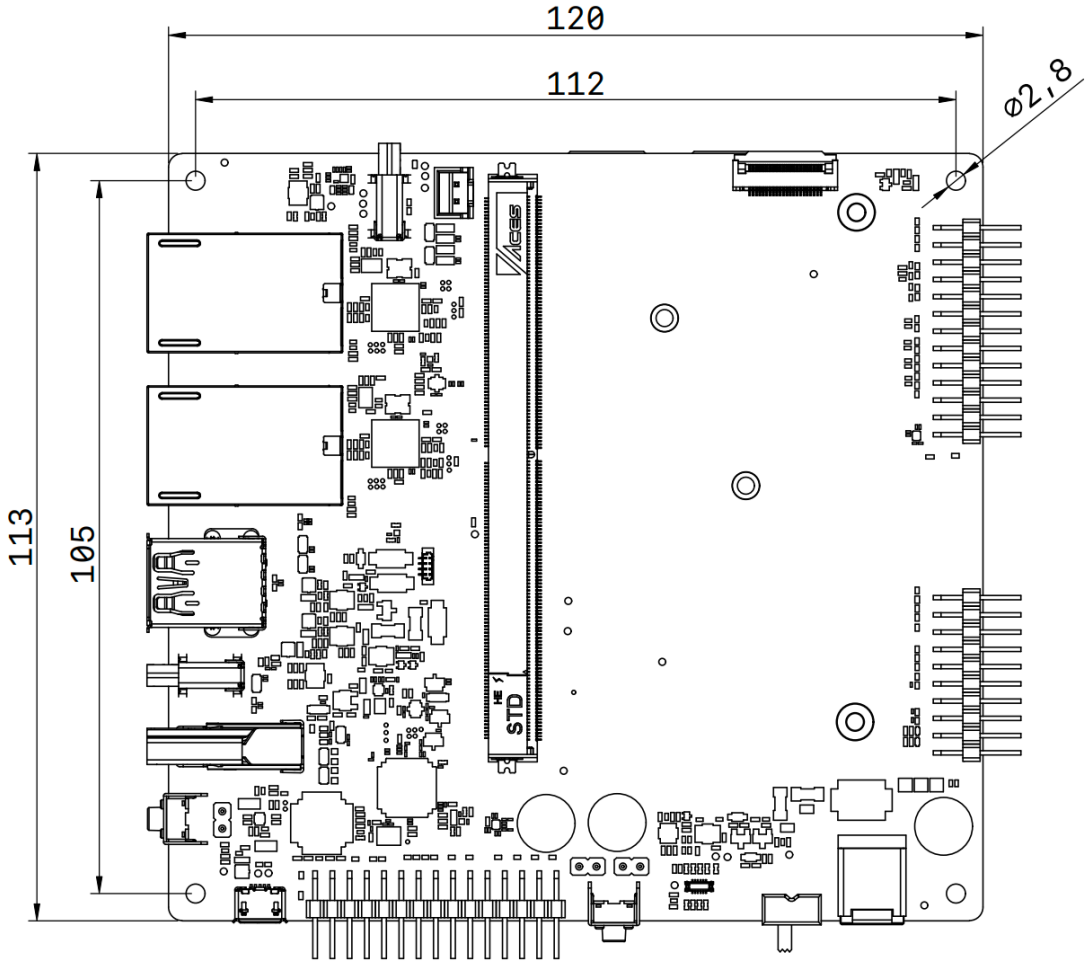


Figure 3. Aura Carrier Board Mechanical Dimensions

3 Carrier Board Connectors Placement

J2103	Aura Core Board Conn.(314-pin)	**J1803**	USB Type-C
J1802	USB Type A Dual Stacked Connector	**J1501**	USB Type-C
J0101	Power Jack	**J0902**	30-pin Expansion Header
J0601	HDMI Connector	**J2001**	Micro-B UART to USB Header
J1201	RJ45 Ethernet Socket	**J0702**	Display Connector
J0801	26-pin Expansion Header	**J1202**	RJ45 Ethernet Socket
J0901	20-pin Expansion Header	**J1402**	Camera1 Connector
J1401	Camera0 Connector	**J1901**	M.2 Key E slot
J0701	LCD Display backlight Power	**J1702**	M.2 Key M slot
J0102	Power ON/Standby Jumper	**S2002**	Wakeup Key
S2001	Reset Key	**S0101**	Main Power Key
J0401	FAN Header	**J2004**	Wakeup key Header
J2002	Reset Key Header		

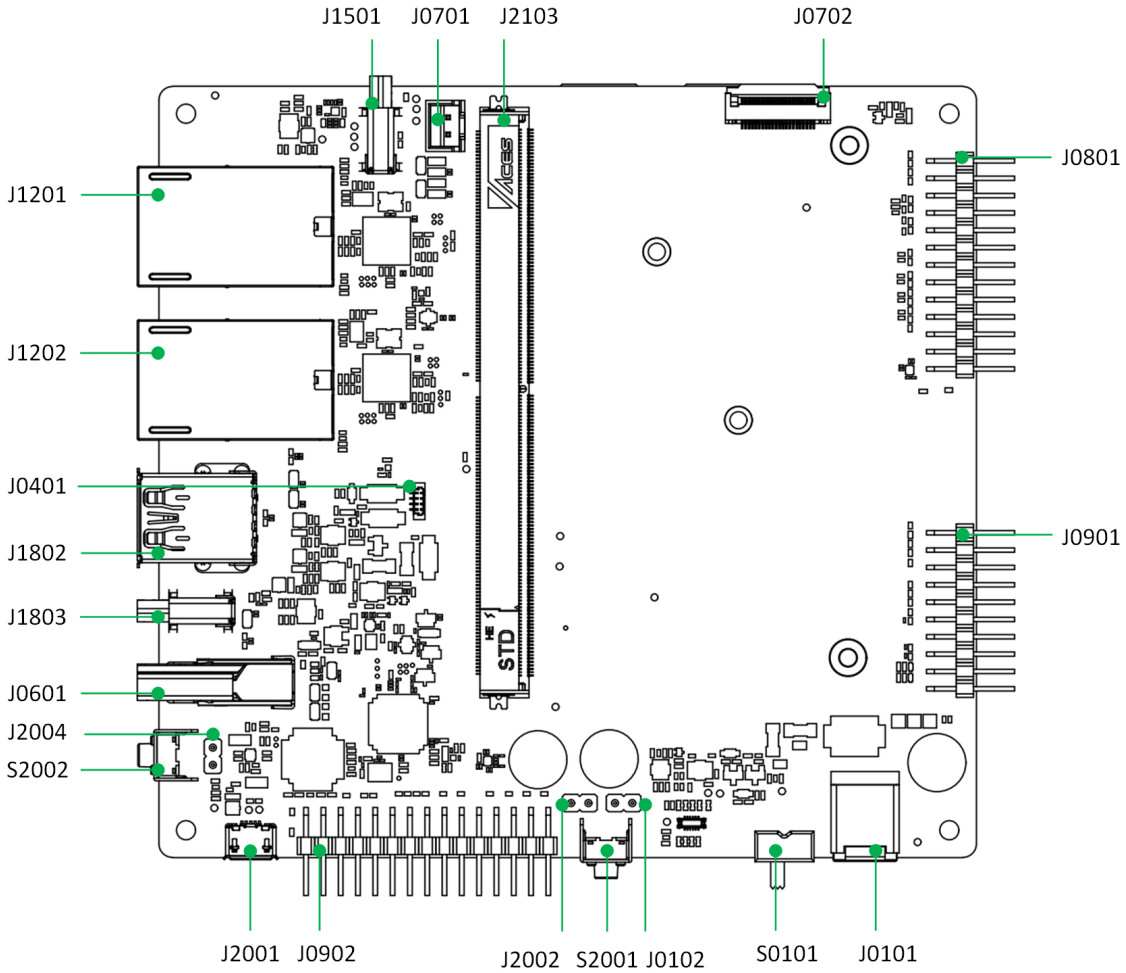


Figure 4. Aura Carrier Board TOP side

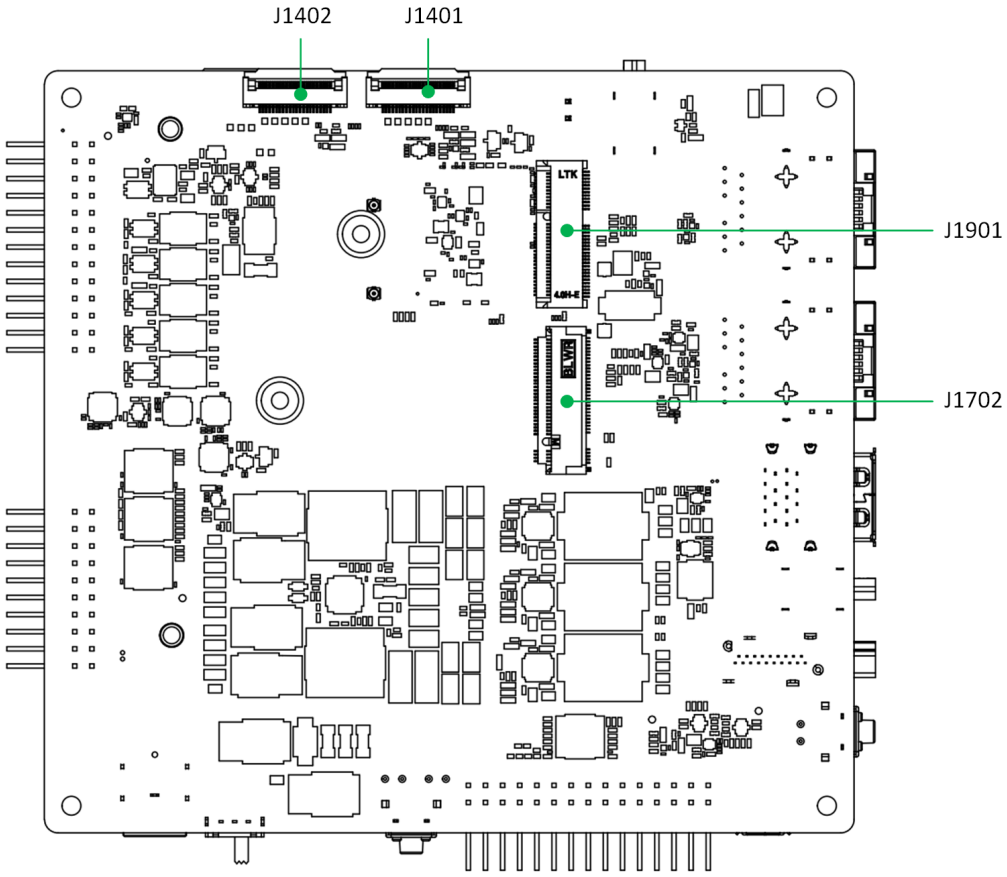


Figure 5. Aura Carrier Board BOT side

4 Board Standard Connectors

Aura carrier board provides several connectors with industry standard pinouts to support additional functions. This includes:

- *2x USB Type-A USB3.1*
- *1x USB Type-C USB3.1*
- *1x USB Type-C USB2.0*
- *1x USB Micro-B*
- *2 x Ethernet RJ45 connector*
- *1x HDMI*
- *M.2 Key M socket*
- *M.2 Key E socket*

4.1 USB Port

The carrier board supports five USB connectors. J1803,J1501 USB Type C connector supporting host, device. J2001 USB Micro-B connector only for download. In addition, there is a dual stacked USB 3.1 Type A connector(J1802). Each Type A connector supports host mode only.

- *2x USB Type-A USB3.1*
- *1x USB Type-C USB3.1 with OTG ,*
- *1x USB Type-C USB host*
- *1x USB Micro-B : Device programming & debug only*

.

Table1 USB Type-A Connector Pin Descriptions – J1802

Pin NO.	Pin Name	——	Usage/Description	Type/Dir
Type A 1#				
1	RST_5V_PORT2		VBUS Supply	Power
2	90OHM_DSP2_DM		USB 2.0 Data from hub	Bidir
3	90OHM_DSP2_DP			
4	GND		Ground	Ground
5	100OHM_DSP2_SS_RXM		USB3.1 Receiver Data from Hub	Input
6	100OHM_DSP2_SS_RXP			
7	GND		Ground	Ground
8	100OHM_DSP2_SS_TXM		USB3.1 Transmit Data from Hub	Output
9	100OHM_DSP2_SS_TXP			
Type A 2#				
10	RST_5V_PORT3		VBUS Supply	Power
11	90OHM_DSP3_DM		USB 2.0 Data from hub	Bidir
12	90OHM_DSP3_DP			
13	GND		Ground	Ground
14	100OHM_DSP3_SS_RXM		USB3.1 Receiver Data from Hub	Input
15	100OHM_DSP3_SS_RXP			
16	GND		Ground	Ground
17	100OHM_DSP3_SS_TXM		USB3.1 Transmit Data from Hub	Output
18	100OHM_DSP3_SS_TXP			

Table 2 USB Type-C Connector Pin Descriptions – J1803

Pin NO.	Pin Name	——	Usage/Description	Type/Dir
A1	GND			
A2	~~100OHM_DSP1_SS_TXP~~		~~USB3.1 Transmit Data from Hub~~	~~Output~~
A3	~~100OHM_DSP1_SS_TXM~~			
A4	RST_5V_PORT1		VBUS Supply	Power
A5	TYPEA_CC1		CC1 from CC Controller	Output
A6	90OHM_DSP1_DP		USB 2.0 Data from hub	Bidir
A7	90OHM_DSP1_DM			
A8	/		Unconnected	
A9	RST_5V_PORT1		BUS Supply	Power
A10	~~100OHM_DSP1_SS_RXM~~		~~USB3.1 Receiver Data from Hub~~	~~Input~~
A11	~~100OHM_DSP1_SS_RXP~~			
A12	GND			
B12	GND			
B11	~~100OHM_DSP1_SS_RXP1~~		~~USB3.1 Receiver Data from Hub~~	~~Input~~
B10	~~100OHM_DSP1_SS_RXM1~~			
B9	RST_5V_PORT1		VBUS Supply	Power
B8	/		Unconnected	
B7	90OHM_DSP1_DM		USB 2.0 Data from hub	Bidir
B6	90OHM_DSP1_DP			
B5	TYPEA_CC2		CC2from CC Controller	Output
B4	RST_5V_PORT1		BUS Supply	Power
B3	~~100OHM_DSP1_SS_TXM1~~		~~USB3.1 Transmit Data from Hub~~	~~Output~~

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B2	~~100OHM_DSP1_SS_TXP1~~			
B1	GND			

Table 3 USB Type-C Connector Pin Descriptions – J1501

Pin NO.	Pin Name	——	Usage/Description	Type/Dir
A1	GND			
A2	100OHM_USB0_TX0_P		USB3.1 Transmit Data from USB0	Output
A3	100OHM_USB0_TX0_N			
A4	USB0_VBUS	USB0_VBUS_CON	VBUS Supply	Power
A5	TYPEA_CC1		CC1 from CC Controller	Output
A6	90OHM_USB0_DP		USB 2.0 Data from hub	Bidir
A7	90OHM_USB0_DN			
A8	/		Unconnected	
A9	USB0_VBUS		BUS Supply	Power
A10	100OHM_USB0_RX1_N		USB3.1 Receiver Data from USB0	Input
A11	100OHM_USB0_RX1_P			
A12	GND			
B12	GND			
B11	100OHM_USB0_RX0_P		USB3.1 Receiver Data from Hub	Input
B10	100OHM_USB0_RX0_M			
B9	USB0_VBUS		VBUS Supply	Power
B8	/		Unconnected	
B7	90OHM_USB0_DN		USB 2.0 Data from hub	Bidir
B6	90OHM_USB0_DP			
B5	TYPEA_CC2		CC2 from CC Controller	Output
B4	USB0_VBUS		BUS Supply	Power
B3	100OHM_USB0_TX1_N		USB3.1 Data Transmit from Hub	Output
B2	100OHM_USB0_TX1_P			
B1	GND			

Table 4 USB Micro-B Connector Pin Descriptions – J2001 (Only Device mode for debug)

Pin NO.	Pin Name	——	Usage/Description	Type/Dir
1	USB_VBUS0			Input
2	90OHM_DEBUG_USB_DP0		Debug UART from CH9104 to ST_DEUART_RXD/TXD	Bidir
3	90OHM_DEBUG_USB_DM0			
4	/		Unconnected	
5	GND		Ground	

4.2 10GbE Ethernet

The carrier board supports two RJ45 connectors for 10G gigabit ethernet (J1201/J1202).

Table 5 Ethernet RJ45 Pin Description – J1201

Pin NO.	Pin Name	——	Usage/Description	Device SCH Side
1	TRCT2		Connect CAP to GND	
2	100OHM_TXVP_B1			

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3	100OHM_TXVN_B1			
4	100OHM_TXVP_C1			
5	100OHM_TXVN_C1			
6	TRCT3		Connect CAP to GND	
7	TRCT1		Connect CAP to GND	
8	100OHM_TXVP_A1			
9	100OHM_TXVN_A1			
10	100OHM_TXVP_D1			
11	100OHM_TXVN_D1			
12	TRCT4		Connect CAP to GND	
13	/			
14	/			
15	/			
16	LED0_1		Green LED-	
17	VCC_SYS_3V3_A		Green LED+	
18	LED1_1		Yellow LED-	
19	VCC_SYS_3V3_A		Yellow LED+	
20	GND		GND	
21	GND		GND	

Pin NO.	**Pin Name**	——	**Usage/Description**	**Cable Side(T568B)**
1	TRP1+		White-Orange	/
2	TRP1-		Orange	/
3	TRP2+		White-Green	/
4	TRP3+		Blue	/
5	TRP3-		White-Blue	/
6	TRP2-		Green	/
7	TRP4+		White-Brown	/
8	TRP4-		Brown	/

Table 6 Ethernet RJ45 Pin Description – J1202

Pin NO.	**Pin Name**	——	**Usage/Description**	**Device SCH Side**
1	TRCT2		Connect CAP to GND	
2	100OHM_TXVP_B2			
3	100OHM_TXVN_B2			
4	100OHM_TXVP_C2			
5	100OHM_TXVN_C2			
6	TRCT3		Connect CAP to GND	
7	TRCT1		Connect CAP to GND	
8	100OHM_TXVP_A2			
9	100OHM_TXVN_A2			
10	100OHM_TXVP_D2			
11	100OHM_TXVN_D2			
12	TRCT4		Connect CAP to GND	
13	/			
14	/			
15	/			

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16	LED0_1		Green LED-	
17	VCC_SYS_3V3_A		Green LED+	
18	LED1_1		Yellow LED-	
19	VCC_SYS_3V3_A		Yellow LED+	
20	GND		GND	
21	GND		GND	

Pin NO.	**Pin Name**	——	**Usage/Description**	**Cable Side(T568B)**
1	TRP1+		White-Orange	/
2	TRP1-		Orange	/
3	TRP2+		White-Green	/
4	TRP3+		Blue	/
5	TRP3-		White-Blue	/
6	TRP2-		Green	/
7	TRP4+		White-Brown	/
8	TRP4-		Brown	/

4.3 HDMI

The carrier board includes one HDMI 2.0 Port(J0601). Output supports data rate up to 6Gbps which provides sufficient bandwidth for 4k@60Hz video.

Table 7 HDMI Port Description – J0601

Pin NO.	**Pin Name**	**Usage/Description**	**Pin NO.**	**Pin Name**	**Usage/Description**	**Type/Dir**
1	100OHM_HDMI_TX2_P_1	TMDS Data2	2	GND		
3	100OHM_HDMI_TX2_N_1		4	100OHM_HDMI_TX1_P_1	TMDS Data1	
5	GND		6	100OHM_HDMI_TX1_N_1		
7	100OHM_HDMI_TX0_P_1	TMDS Data0	8	GND		
9	100OHM_HDMI_TX0_N_1		10	100OHM_HDMI_CLK_P_1	TMDS clock	
11	GND		12	100OHM_HDMI_CLK_N_1		
13	CEC_HDMI	Consumer Electronic control	14	/		
15	SCL_HDMI		16	SDA_HDMI		
17	GND		18	HDMI_5V	Power Supply	
19	DETECT_HDMI	HDMI Hotplug				

4.4 M.2 Key M Slot

The carrier board includes one M.2, Key M Slots for NVMe storage (J1702). The M.2, Key M slot supports PCIe (x2), Gen4.

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Table 8 M.2 Key M Slot Pin Description – J1702

Pin NO.	Pin Name	Usage/Description	Pin NO.	Pin Name	Usage/Description	Type/Dir
1	GND		2	VCC_3V3_NVMe		
3	GND		4	VCC_3V3_NVMe		
5	/		6	/		
7	/		8	/		
9	GND		10	LED#		3.3V
11	/		12	VCC_3V3_NVMe		
13	/		14	VCC_3V3_NVMe		
15	GND		16	VCC_3V3_NVMe		
17	/		18	VCC_3V3_NVMe		
19	/		20	/		
21	GND		22	/		
23	/		24	/		
25	/		26	/		
27	GND		28	/		
29	85OHM_PCIE1_RX1_N	PCIE lane1 Receive	30	/		
31	85OHM_PCIE1_RX1_P		32	/		
33	GND		34	/		
35	85OHM_PCIE1_TX1_N	PCIE lane1 Transmit	36	/		
37	85OHM_PCIE1_TX1_P		38	/		
39	GND		40	/		
41	85OHM_PCIE1_RX0_N	PCIE lane0 Receive	42	/		
43	85OHM_PCIE1_RX0_P		44	WK_GPIO34	Key M alert	1.8V
45	GND		46	/		
47	85OHM_PCIE1_TX0_N	PCIE lane0 Transmit	48	/		
49	85OHM_PCIE1_TX0_P		50	PHY_RESET_N_3V3	PCIE Reset	3.3V
51	GND		52	PHY_PCIE_CLKREQB_V_3V3	PCIE Clock Request	IN 3.3V
53	85OHM_PCIE1_CLK_N	PCIE ref CLK	54	PCIE1_WAKE_N_3V3	PCIE wake	IN
55	85OHM_PCIE1_CLK_P		56	/		
57	GND		58	/		
59	KEY		60	KEY		Unused
61			62			
63			64			
65			66			
67	/		68	/		
69	/		70	VCC_3V3_NVMe		
71	GND		72	VCC_3V3_NVMe		
73	GND		74	VCC_3V3_NVMe		
75	GND					

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4.5 M.2 Key E Slot

The carrier board includes one M.2, Key E Slots for Wireless network card (Wi-Fi), Bluetooth, WWAN (Mobile Network) module (J1901). The M.2, Key E slot supports PCIe (x1), Gen4.

Table 9 M.2 Key E Slot Pin Description – J1901

Pin NO.	Pin Name	Usage/Description	Pin NO.	Pin Name	Usage/Description	Type/Dir
1	GND					
3	/		2	VCC		
5	/		4	VCC		
7	GND		6	/		
9	/		8	SC_PCM3_CLK		1.8V
11	/		10	SC_PCM3_SYNC		1.8V
13	/		12	SC_PCM3_IN		1.8V
15	/		14	SC_PCM3_OUT		1.8V
17	/		16	/		
19	/		18	GND		
21	/		20	BT_WAKE_HOST_3V3	Bluetooth Wake AP	IN, 3.3V
23	/		22	SC_UART3_RXD		1.8V
25	KEY		24	KEY		Unused
27			26			
29			28			
31			30			
33	GND		32	SC_UART3_TXD		1.8V
35	85OHM_PCIE0_TX0_P	PCIE Transmit Lane0	34	SC_UART3_CTS		1.8V
37	85OHM_PCIE0_TX0_N		36	SC_UART3_RTS		1.8V
39	GND		38	/		
41	85OHM_PCIE0_RX0_P	PCIE Receive Lane0	40	/		
43	85OHM_PCIE0_RX0_N		42	/		
45	GND		44	/		
47	85OHM_PCIE0_X4_REF_CLK_P	PCIE Reference Clock	46	/		
49	85OHM_PCIE0_X4_REF_CLK_N		48	HOST_WAKE_BT		
51	GND		50	/		
53	PCIE_CLKREQB_N_3V3	Clock Request	52	PCIE_RESET_N_3V3	PCIE Reset	3.3V
55	PCIE_WAKE_N_3V3	PCIE Wake	54	BT_EN_3V3	BT enable	3.3V
57	GND		56	WLAN_EN_3V3	WIFI enable	3.3V
59	/		58	/		
61	/		60	/		
63	GND		62	/		
65	/		64	/		
67	/		66	/		
69	GND		68	/		
71	/		70	/		
73	/		72	VCC		
75	GND		74	VCC		

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5 Board Expansion IF Connectors

The Aura carrier board supports several expansion headers and connectors that have custom pinouts. The following lists the headers and connectors that have custom pinouts.

- *Aura Module Connector, 314-pin, MxM + 12-pin B2B connector*
- *MIPI Display Connectors (x1), 22 position, Flex Connector, 0.5 mm pitch*
- *MIPI Camera Connectors (x2), 22 position, Flex Connector, 0.5 mm pitch*
- *30-Pin Expansion Header, 2x15, 2.54 mm pitch*
- *26-Pin Expansion Header, 2x13, 2.54 mm pitch*
- *20-Pin Expansion Header, 2x10, 2.54 mm pitch*
- *Fan Connector, 4-pin, 1.25 mm pitch*
- *DC Power Jack*

5.1 Carrier Board 30+26+20 pin expansion header

The Carrier Board includes a 30-pin (2x15, 2.54 mm pitch) J0902 ,a 26-pin (2x13, 2.54 mm pitch)expansion header (J0801) and a 20-pin (2x10, 2.54 mm pitch)expansion J0901. The expansion connector includes various control interfaces including:

6x CANFD
2x SPI:SC_SPI0,SC_SPI1
2x ADC:ADC_IN0,ADC_IN1
5x UART: SC_UART2(1.8V& 3.3V),RT_UART0/ 1/2/3
4x PWM:PWM_OUT5,PWM_OUT6,PWM_OUT7,PWM_OUT8
1x I2S:SC_I2S0
3x I2C:SC_I2C4,SC_I2C5,SC_I2C6
1.8V
3.3V
5V

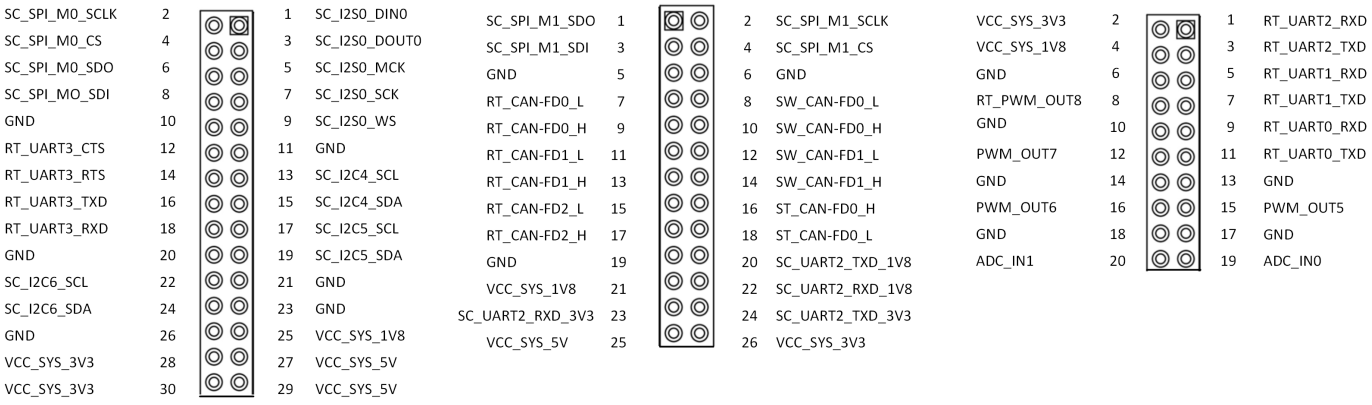


Figure 6 Aura Carrier Board Expansion Header Connectors
(J0902 / J0801 / J0901)

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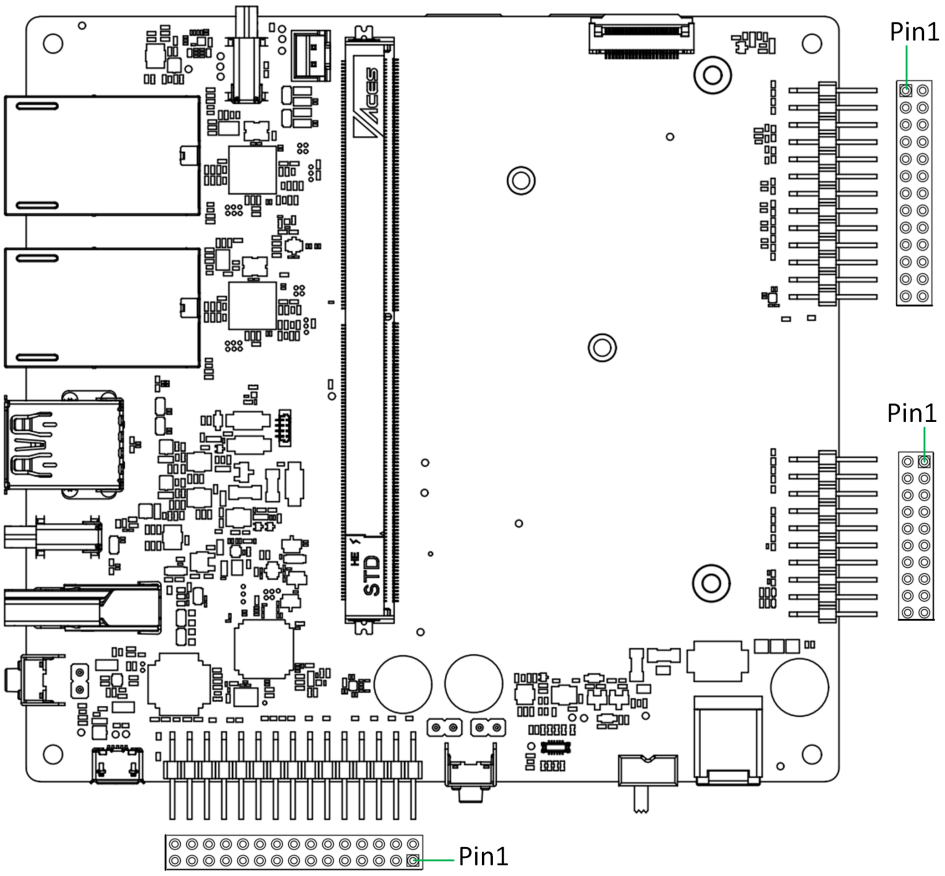


Figure7 Expansion Header J0901/J0902/J0801 Connectors PIN1 direction

Comments:

1. For all GPIOs, the default voltage range is 3.3V. It will be marked on the network,which is a 1.8V GPIO.
2. ADC Ports: The maximum input voltage of the ADC port is 1.8V
3. PWM Ports:All the PWM on carrier belong to the RT control domain.

5.2 Carrier Board CSI MIPI 22 pin Connector

The Carrier Board includes two 22-position flex (0.5 mm pitch) camera connectors (J1402 and J1401). The connector used on the carrier board Part #FPC05022-09204.

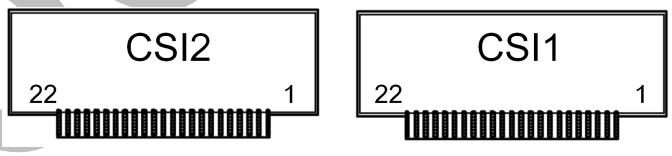


Figure 8 Aura Carrier Board Camera MIPI Connectors Pin1 Placement

Table 10 Carrier Board Carrier Board CSI pin assignments - J1401

Pin NO.	Pin Name	——	Usage/Description	Type/Dir
1	VCC_CSI_3V3_1		VDD_3.3V	
2	SC_I2C6_SDA_3.3		CAM_I2C_SDA	Bidir
3	SC_I2C6_SCL_3.3		CAM_I2C_SCL	OUTPUT
4	GND		GND	
5	RT_CANFD3_TX (WK_GPIO34)		CAM_MCLK	OUTPUT

6	PWDN0		CAM_PWDN_LS	INPUT
7	GND		GND	
8	100OHM_MIPI_CSI1_RX_D3P_CON		CSI_D3P	INPUT
9	100OHM_MIPI_CSI1_RX_D3N_CON		CSI_D3N	INPUT
10	GND		GND	
11	100OHM_MIPI_CSI1_RX_D2P_CON		CSI_D2P	INPUT
12	100OHM_MIPI_CSI1_RX_D2N_CON		CSI_D2N	INPUT
13	GND		GND	
14	100OHM_MIPI_CSI1_RX_CKP_CON		CSI_CLKP	INPUT
15	100OHM_MIPI_CSI1_RX_CKN_CON		CSI_CLKN	INPUT
16	GND		GND	
17	100OHM_MIPI_CSI1_RX_D1P_CON		CSI_D1P	INPUT
18	100OHM_MIPI_CSI1_RX_D1N_CON		CSI_D1N	INPUT
19	GND		GND	
20	100OHM_MIPI_CSI1_RX_D0P_CON		CSI_D0P	INPUT
21	100OHM_MIPI_CSI1_RX_D0N_CON		CSI_D0N	INPUT
22	GND		GND	

Table 11 Carrier Board Carrier Board CSI pin assignments - J1402

Pin NO.	**Pin Name**	——	**Usage/Description**	**Type/Dir**
1	VCC_CSI_3V3_2		VDD_3.3V	
2	SC_I2C6_SDA_3.3		CAM_I2C_SDA	Bidir
3	SC_I2C6_SCL_3.3		CAM_I2C_SCL	OUTPUT
4	GND		GND	
5	RT_CANFD3_RX (WK_GPIO35)		CAM_MCLK	OUTPUT
6	PWDN1		CAM_PWDN_LS	INPUT
7	GND		GND	
8	100OHM_MIPI_CSI2_RX_D3P_CON		CSI_D3P	INPUT
9	100OHM_MIPI_CSI2_RX_D3N_CON		CSI_D3N	INPUT
10	GND		GND	
11	100OHM_MIPI_CSI2_RX_D2P_CON		CSI_D2P	INPUT
12	100OHM_MIPI_CSI2_RX_D2N_CON		CSI_D2N	INPUT
13	GND		GND	
14	100OHM_MIPI_CSI2_RX_CKP_CON		CSI_CLKP	INPUT
15	100OHM_MIPI_CSI2_RX_CKN_CON		CSI_CLKN	INPUT
16	GND		GND	
17	100OHM_MIPI_CSI2_RX_D1P_CON		CSI_D1P	INPUT
18	100OHM_MIPI_CSI2_RX_D1N_CON		CSI_D1N	INPUT
19	GND		GND	
20	100OHM_MIPI_CSI2_RX_D0P_CON		CSI_D0P	INPUT
21	100OHM_MIPI_CSI2_RX_D0N_CON		CSI_D0N	INPUT
22	GND		GND	

5.3 Carrier Board DSI MIPI 22 pin Connector

The Carrier Board provides a 22-position flex (0.5 mm pitch) connector for MIPI Display (J0702). The connector used on the carrier board is FPC05022-09204

Osanex



Figure 9 Aura Carrier Board Display MIPI Connector Pin1 Placement

Table 12 Carrier Board Carrier Board DSI pin assignments - J0702

Pin NO.	Pin Name	——	Usage/Description	Type/Dir
1	VCC_DSI_3V3		3.3V Power Supply	
2	LVDS_TX_SDA		SDA	OUTPUT
3	LVDS_TX_SCL		SCL	OUTPUT
4	GND		GND	
5	/		NC	
6	/		NC	
7	GND		GND	
8	100OHM_MIPI_DSI1_D3P_CON		DSI_D3P	OUTPUT
9	100OHM_MIPI_DSI1_D3N_CON		DSI_D3N	OUTPUT
10	GND		GND	
11	100OHM_MIPI_DSI1_D2P_CON		DSI_D2P	OUTPUT
12	100OHM_MIPI_DSI1_D2N_CON		DSI_D2N	OUTPUT
13	GND		GND	
14	100OHM_MIPI_DSI1_CKP_CON		DSI_CLKP	OUTPUT
15	100OHM_MIPI_DSI1_CKN_CON		DSI_CLKN	OUTPUT
16	GND		GND	
17	100OHM_MIPI_DSI1_D1P_CON		DSI_D1P	OUTPUT
18	100OHM_MIPI_DSI1_D1N_CON		DSI_D1N	OUTPUT
19	GND		GND	
20	100OHM_MIPI_DSI1_D0P_CON		DSI_D0P	OUTPUT
21	100OHM_MIPI_DSI1_D0N_CON		DSI_D0N	OUTPUT
22	GND		GND	

5.4 Fan Connector

Table 13 The Carrier board includes a 4-pin fan header (J0401).

Pin NO.	Pin Name	——	Usage/Description	Type/Dir
1	SW_FRAY0_A_RXDSC_PWM0		PWM FAN CONTROL	OUTPUT
2	ST_GPIO16		FAN TACH indicate	INPUT
3	VCC_SYS_12V		FAN POWER	
4	GND		GND	

5.5 Power ON/Standby Jumper

Table 14 The Carrier board includes a Standby Jumper (J0102).

Pin NO.	Pin Name	Usage/Description
1	GND	
2	Vout	Default controled by power on sequence, short to GND Standby Aura module

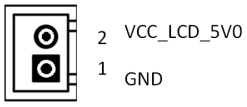
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5.6 LCD Display Backlight Header

Table 15 The Carrier board LCD backlight Header(J0701)

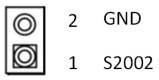
Pin NO.	Pin Name	Usage/Description
1	GND	
2	VCC_LCD_5V0	5V power output for LCD backlight power supply



5.7 Wakeup Key Header Jumper

Table 16 The Carrier board includes a Wakeup key Header Jumper (J2004).

Pin NO.	Pin Name	Usage/Description
1	S2002	Connect to KEY S2002 for Wakeup key extension cord
2	GND	



5.8 Rest Key Header Jumper

Table 17 The Carrier board includes a Rest Key Header Jumper (J2002).

Pin NO.	Pin Name	Usage/Description
1	S2001	Connect to KEY S2001 for REST key extension cord
2	GND	



5.9 DC Power Jack

The Aura carrier board uses a DC power jack (J0101) to bring in the power from the included DC power supply. The mating barrel jack connector dimensions.

Length: 9.5 mm
Diameter1: 5.5 mm
Diameter2: 2.5mm
The center pin is positive (+V)
Max current supported is 10A

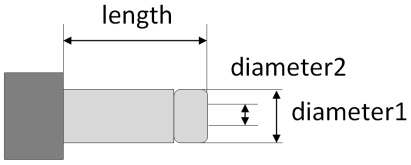


Figure 10 DC-IN Plug (DC-2.5-PU)

Seeed

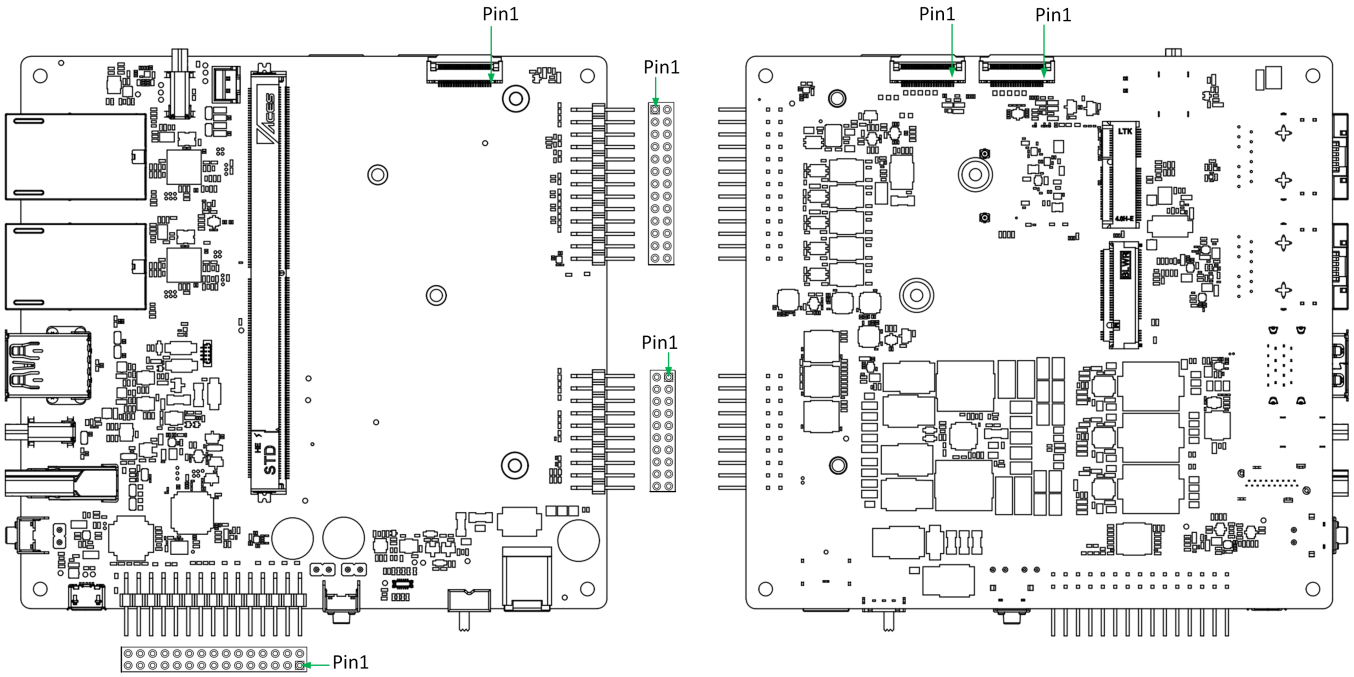


Figure 11 Carrier Board expension connectors PIN 1

Sesame

6 Power Supplies

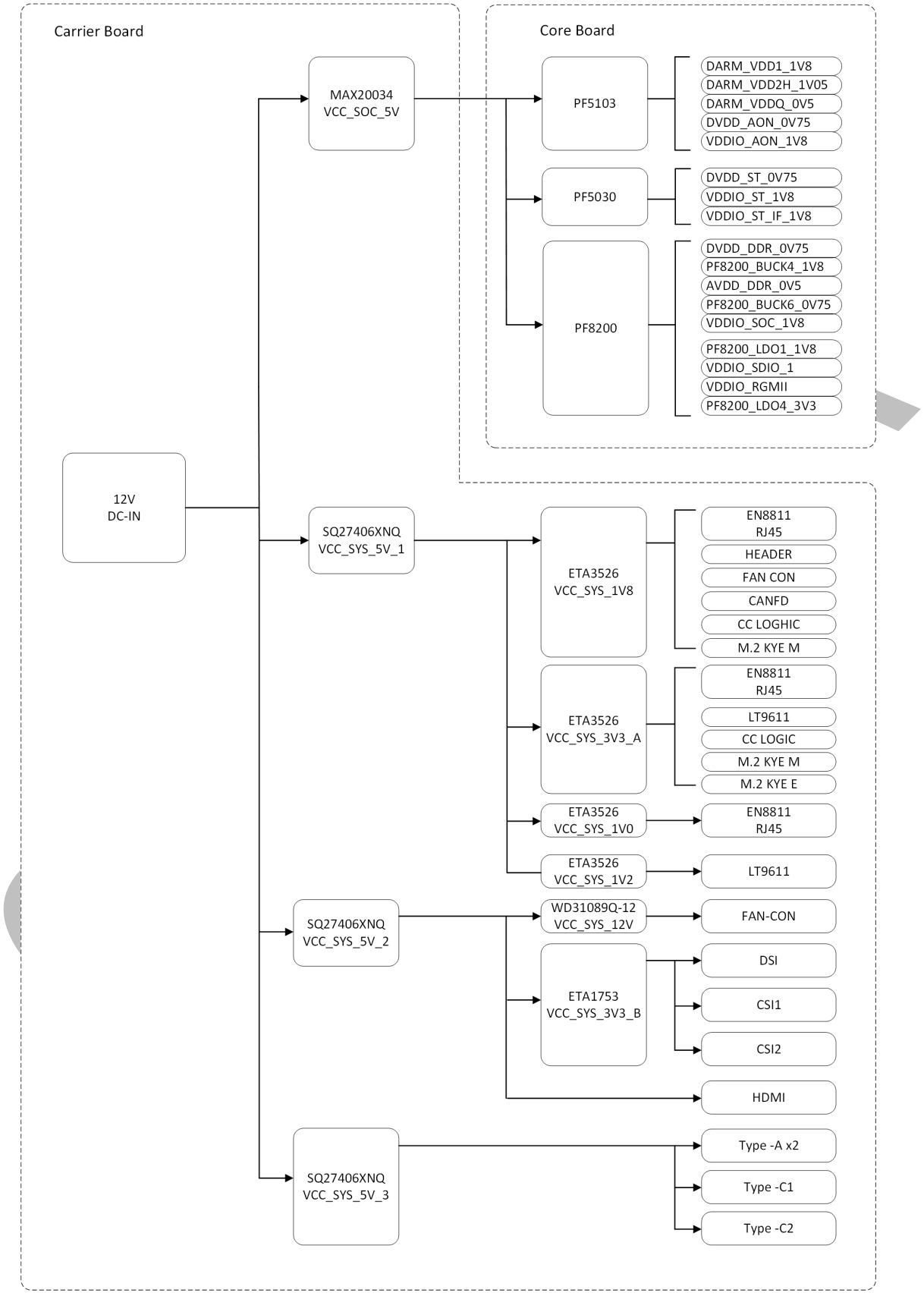


Figure 12 Aura Core Board & Carrier Board Power Diagram

The following tables show the allocation of supplies to the connectors on the carrier board and current capabilities.

Table 18 Interface Power Supply Allocation

Power	Usage	V	Power Supply or Gate	Power Source	Enable
VCC_PWR_IN	Main Power DC-IN Adapter	12~20V			
VCC_SYS_5V0_1	1# Main 5V	5	SQ27406XNQ	**VCC_PWR_IN**	Enable IC
VCC_SYS_5V0_2	2# Main 5V & PIN HEADER	5	SQ27406XNQ	**VCC_PWR_IN**	Enable IC
VCC_SYS_5V0_3	3# Main 5V	5	SQ27406XNQ	**VCC_PWR_IN**	Enable IC
VCC_SYS_3V3_A	A# Main 3.3V	3.3	ETA3526FSG	**VCC_SYS_5V0_1**	Enable IC
VCC_SYS_3V3_B	B# Main 3.3V & PIN HEADER	3.3	ETA3526FSG	**VCC_SYS_5V0_2**	Enable IC
VCC_SYS_1V8	30/26/20 PIN HEADER	1.8	ETA3526FSG	**VCC_SYS_5V0_1**	Enable IC
RST_5V_PORT3	USB-A 2 J1802 VBUS POWER	5	SGM2588DYN5G/TR	**VCC_SYS_5V0_3**	USB Hub
RST_5V_PORT2	USB-A 1 J1802 VBUS POWER	5	SGM2588DYN5G/TR	**VCC_SYS_5V0_3**	USB Hub
RST_5V_PORT1	USB-C J1803 VBUS POWER	5	SGM2588DYN5G/TR	**VCC_SYS_5V0_3**	USB Hub
USB0_VBUS	USB-C J1501 VBUS POWER	5	SGM2588DYN5G/TR	**VCC_SYS_5V0_3**	Enable IC
USB_VBUS0	USB-MicroB VBUS POWER	5	/	From PC host	
+3.3V_BD_1	Ethernet J1201 LED power	3.3	R1036	**VCC_SYS_3V3_A**	LAN1
+3.3V_BD_2	Ethernet J1202 LED power	3.3	R1138	**VCC_SYS_3V3_A**	LAN2
HDMI_5V	HDMI POWER	5	SGM2588DYN5G/TR	**VCC_SYS_5V0_2**	HDMI TX
VCC_3V3_NVMe	M.2 KEY M POWER	3.3	SGM2575ADYG/TR	**VCC_SYS_3V3_B**	M.2 K M
VCC	M.2 KEY E POWER	3.3	SGM2575ADYG/TR	**VCC_SYS_3V3_A**	M.2 K E
VCC_CSI_3V3_1	CSI CAMERA CONNECTOR	3.3	SGM2575ADYG/TR	**VCC_SYS_3V3_B**	Enable IC
VCC_CSI_3V3_2	CSI CAMERA CONNECTOR	3.3	SGM2575ADYG/TR	**VCC_SYS_3V3_B**	Enable IC
VCC_DSI_3V3	DSI DISPLAY CONNECTOR	3.3	WPM3027-3/TR	**VCC_SYS_3V3_B**	

Table 19 Supply Current Capabilities per Connector

Connectors	Pin NO.	Voltage (V)	Max Current (A)	Usage	Power Net	Total Power (W)
J2103	1~18	5	13	Core board 5V Power supply	VCC_SYS_5V0	65
J0801	25	5	1	26 PIN Header 5V	VCC_SYS_5V0_2	10
J0902	27/29	5	1	30 PIN Header 5V		
J0801	26	3.3	0.5	26 PIN Header 3.3V	VCC_SYS_3V3_B	4.95
J0901	2	3.3	0.5	20 PIN Header 3.3V		
J0902	28/30	3.3	0.5	30 PIN Header 3.3V		
J0801	24	1.8	0.5	26 PIN Header 1.8V	VCC_SYS_1V8	2.7
J0901	4	1.8	0.5	20 PIN Header 1.8V		
J0902	25	1.8	0.5	30 PIN Header 1.8V		
J0702	1	3.3	0.35	MIPI Display Module Power	VCC_DSI_3V3	1.155
J0701	2	5	0.25	LCD Display Backlight Power	VCC_LCD_5V0	1.25
J1401	1	3.3	0.7	MIPI Camera 1 Module Power	VCC_CSI_3V3_1	2.31
J1402	1	3.3	0.7	MIPI Camera 2 Module Power	VCC_CSI_3V3_2	2.31
J1501	A4/A9/B4/B9	5	2	USB TypeC Port Vbus Power	USB0_VBUS_CON	10
J1802	1	5	2	USB Type A Port 1# Vbus	RST_5V_PORT2	10
	10	5	2	USB Type A Port 2# Vbus	RST_5V_PORT3	10
J1803	A4/A9/B4/B9	5	2	USB TypeC Port Vbus Power	RST_5V_PORT1	10

Sample

J0601	18	5	1	HDMI Display Port Power	HDMI_5V	5
J1901	2/4/72/74	3.3	3	PCIE M.2 BT/WIFI Power	M.2 Key E	9.9
J1702	2/4/12/14/16 /18/70/72/74	3.3	2	PCIE M.2 SSD Power	VCC_3V3_NVMe	6.6

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7 Device Ordering Information

DX-5000-D	Module Form Factor : MXM	
DX-5000-D008	Memory Config: 8GB DRAM	
DX-5000-D016	Memory Config: 16GB DRAM	

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